

Sriharsh Bhyravajjula

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EDUCATION

University of Washington

Seattle, USA

Master of Science (Computational Linguistics), GPA: 3.9/4.0

Sep. 2022 – Mar. 2024

Relevant Courses: Deep Learning, Machine Learning for NLP, Syntax Engineering, Shallow and Deep Techniques in NLP

International Institute of Information Technology

Hyderabad, India

Bachelor of Science (Computer Science), GPA: 3.2/4.0

Aug. 2014 – Aug. 2018

Relevant Courses: Data Structures, Algorithms, Statistical Methods in AI, Natural Language Processing, Speech Systems

- **Dean's Merit List Awardee** for exceptional academic performance — Monsoon 2017, Spring 2018

EXPERIENCE

Outreach Corporation

Seattle, Washington

Senior Applied Scientist

Oct. 2023 – current

- Building an **AI agent evaluation framework** for Outreach's LLM tools ecosystem by defining error taxonomies (like hallucinated params and sequencing failures) and generating synthetic tool-use scenarios to ensure **MCP-server compatibility** and reliable **tool invocation at scale**.
- Developing **hierarchical summarization** for **account/deal intelligence** at varied levels of abstraction by combining incremental chronological updates with multi-modal fusion (transcripts, emails, CRM signals) to surface high-level and actionable sales insights.
- Built an end-to-end **LLM evaluation** platform processing 90K meeting transcripts+summaries to monitor weekly **summarization quality** with MLflow tracing for LLM observability. Added perturbation-based sensitivity testing across 20+ semantic axes to validate evaluation robustness, revealing and fixing **systematic evaluator blind spots** for **LLM-as-a-judge/human alignment**. Reduced eval time from 3-5 manual weeks to 2hr job runs.
- Improved Outreach's **retrieval-augmented generation (RAG)** pipeline built for sales-facing QA by 15-30% on retrieval (nDCG/P@k), and 20-25% on answer generation (f1) through controlled ablations across contextual retrieval, multi-vector embeddings, BM25+semantic hybrid approaches, temporal reranking, and query decomposition strategies, with documented zero-downtime migration strategies.
- **Won Outreach 2024 Hackathon** (Novel AI category) by building a predictive deal-risk agent. Owned Azure OpenAI deployments(\$100K+ monthly), k8s infra, and Datadog observability during **on-call rotations**.

Applied Science Intern

Jun. 2023 – Sep. 2023

- Used in-context instruction learning techniques in **prompt tuning** experiments leveraging LLMs to summarize answers to sales methodology questions from meeting transcripts.

Klevu AI

Hyderabad, India

NLP Data Scientist II

Jan. 2021 – Aug. 2022

- Built Klevu's ecommerce **knowledge graph** to semantically understand and answer complex search queries by creating APIs which curate information from domain-rich sources (product reviews, knowledge-bases, user analytics, and catalogs), and pool this data into custom typeDB databases capable of real-time inference.
- Analysed **product reviews** to enhance Klevu's ecommerce catalogs with product use-cases and feature sentiment, providing retailers with premium Business Intelligence (BI). Used a BiLSTM model in a grid-tagging scheme to label **aspect-opinion-sentiment triplets** in reviews with an F1-score of 0.68 across two test domains.
- Reduced **zero-result searches** and improved **search conversion** by building and deploying an end-to-end synonym-enrichment API. Given product-names from store catalogs, the API extracts corresponding synonyms from the Open Multilingual Wordnet, and disambiguates their synset glosses against product descriptions with Transformer-based **embedding models** like multilingual universal sentence encoder and mT5.

- Designed and built backend for Klevu's [Enhanced Search Analytics](#), which featured in [Forbes](#) and helped ecommerce retailers understand shopper behavior and boost sales. This involved creation of (1) a scalable Elasticsearch stack for 1.5B+ data points (clicks, searches, checkouts), (2) NLP/ML scripts extracting BI like **ecommerce conversion metrics**, intuitive **word-graphs for search patterns**, and **time-series predictions** in real-time, and (3) interactive dashboards where these insights were served concurrently for 2000+ ecommerce retailers.

TECHNICAL SKILLS

LLM/Agents: langchain, langgraph, langsmith, huggingface, llamaindex, openai, autogen, MCP, safetensors, GGUF
ML/NLP Frameworks: torch, sklearn, stanza, spacy, tensorflow, keras, mlflow, opentelemetry, pydantic
Programming & Scripting: python, C/C++, java, shell, L^AT_EX
Data Management: pandas, databricks, spark, elasticsearch, sql, postgresSQL, mongoDB, typedb, neo4j
Deployment: azure, gcp, aws, docker, k8s, fastapi, flask, vllm, sglang, terraform, datadog, torchserve
IDE, CI, & Version Control: VS code, sublime, pycharm, jupyterlab, CircleCI, git

PUBLICATIONS

[so much depends / upon / a whitespace: Why Whitespace Matters for Poets and LLMs](#)

Sriharsh Bhyravajjula, Melanie Walsh, Anna Preus and Maria Antoniak — EMNLP (Main) 2025 : Suzhou, China

- Analyzed whitespace as a semantic-spatial element in poetry using 19k Poetry Foundation poems by 4k poets, releasing 2.8k public-domain texts with original formatting. Created WISP, a novel whitespace typology, and released WISP-Bench, a benchmark to evaluate whitespace performance in linearization methods that capture image-to-text, highlighting the impacts of dataset preparation on NLP and LLM tokenization design strategies.

[MARCUS: An Event-Centric NLP Pipeline that generates Character Arcs from Narratives](#)

Sriharsh Bhyravajjula, Ujwal Narayan and Manish Shrivastava — Text2Story @ ECIR 2022 : Stavanger, Norway

- Won the **Best Paper award** at the workshop and was given a **Student Travel Grant** by IIIT-H.
- Built a novel pipeline that (1) used a BiLSTM tagger to label **events** and a Semantic Role Labeller to identify **participant characters** from long-form narratives, (2) used fine-tuned BERT and RoBERTa models to extract proxy markers of emotion and sentiment to measure a **character's circumstance**, and (3) plotted smoothed **character arcs** as an aggregated change of character circumstance across event sequences.

POSITIONS OF RESPONSIBILITY

- 3 x Program Committee Member, 3 x Recognized Reviewer Award:** Text2Story @ ECIR '23, '24, '25
- Reviewer:** ACL ARR October 2025
- Teaching Assistant** for *CSE472: Natural Language Processing* (Monsoon 2017, Monsoon 2018)
- Teaching Assistant** for *HSS345: Introduction to Shakespeare* (Spring 2018) and *HSS445: Literature - American Classics* (Spring 2019) at IIIT-Hyderabad.
- Editor-in-Chief** of IIIT-Hyderabad's student-run college magazine, [Ping!](#) — Aug. 2017 - Aug. 2018
- President** of IIIT-Hyderabad's Literary Club — Aug. 2016 - Aug. 2017
- Invited Speaker** on *Advances in NLP* at [AICC 2018](#), Hyderabad.
- Guest Reporter** to *The Adivasi Academy, Tejjadh*, covering Padma Shri Ganesh Devy's [People's Linguistic Survey of India](#), a nationwide initiative documenting 780 languages across India.

WRITING

- Debut poetry collection: [sorting through paperwork after midnight](#), published by Authorspress, May 2019.
- Poetry and prose published in magazines like [Indian Literature](#) (Issues 275, 286), [Muse India](#) (Issue 65), and [Madras Courier](#) (Issues 75, 85). Shortlisted for the [Wingword Poetry Prize, 2021](#).